

TCLMA: A Two-dimension Contrastive Learning based Multiagent Framework for Scientific Novelty Evaluation

Lin Zheng

Department of Information Systems, School of Business
City University of Hong Kong
Hong Kong, China
linzheng4-c@my.cityu.edu.hk

Jian Ma

Department of Information Systems, School of Business
City University of Hong Kong
Hong Kong, China
isjian@cityu.edu.hk

Qingyu Xu*

Department of Information Systems, School of Business
City University of Hong Kong
Hong Kong, China
qingyuxu4-c@my.cityu.edu.hk

Yong Wang

Department of Economics and Finance
City University of Hong Kong
Hong Kong, China
efywang@cityu.edu.hk

Abstract

Novelty assessment is a cornerstone of technological innovation and intellectual property protection, profoundly influencing national technology strategies and economic development. Traditional evaluation models, which rely on domain experts, face significant challenges, including subjectivity, inconsistency, and low efficiency. These issues are exacerbated by the surging volume of global patent applications. While generative AI technologies, particularly Large Language Models (LLMs), offer promising avenues for automation, their prevalent use of implicit reasoning often leads to "hallucinations" and weak discrimination capabilities, rendering them unsuitable for the logically rigorous task of patent novelty assessment. To address this gap, this paper introduces TCL-MFSNE (Temporal-Contrastive Learning Multi-Agent Framework with Byzantine Consensus for Scientific Novelty Evaluation), an advanced multi-agent framework designed to systematically enhance the accuracy, robustness, and interpretability of evaluations. The core of TCL-MFSNE rests on three pillars: (1) Explicit Chain-of-Thought (CoT) Reasoning, which guides the model to generate detailed, transparent reasoning steps, overcoming the black-box nature of implicit reasoning. (2) Two-Dimensional Contrastive Learning, which includes "Positive-Negative Contrastive Learning" to sharpen the model's ability to discern key novel features by learning from paired authorized and unauthorized patents, and "Temporal Contrastive Learning" to ensure evaluations are accurately contextualized within the historical technological landscape. (3) Byzantine Consensus Mechanism, which leverages multiple heterogeneous LLM agents for independent evaluations and employs a consensus algorithm to filter out individual agent errors or biases, ensuring a stable and reliable final decision. We conducted comprehensive experiments on a real-world dataset of patents filed between 2008 and 2024. The results demonstrate that the TCL-MFSNE framework

achieves an accuracy and F1-score of 85%, significantly outperforming baseline models such as implicit LLMs, single-agent systems, and other ablation variants. This research not only provides a viable solution for automated patent novelty assessment but also offers a new paradigm for applying generative AI to other professional domains that demand high precision and strong logical coherence.

CCS Concepts

• **Information systems** → Information systems applications; Decision support systems.

Keywords

Large Language Models, Novelty Evaluation, Patent Evaluation, Multiagent Systems

ACM Reference Format:

Lin Zheng, Qingyu Xu, Jian Ma, and Yong Wang. 2025. TCLMA: A Two-dimension Contrastive Learning based Multiagent Framework for Scientific Novelty Evaluation. In *2025 International Conference on Generative Artificial Intelligence for Business (GAIB 2025)*, August 04–06, 2025, Hongkong, China. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3766918.3766939>

1 Introduction

In the era of globalization and the knowledge economy, technological innovation serves as the core engine driving social progress and national competitiveness. The patent system, as the bedrock for protecting innovative achievements and incentivizing R&D investment, has never been more critical. Within this system, novelty assessment stands as the most fundamental yet challenging step in the patent granting process. According to patent law, an invention must possess novelty, meaning that before its filing date, the same invention has not been publicly disclosed in publications, used publicly, or otherwise made known to the public [1]. This process essentially involves a comprehensive and meticulous comparison of the claimed invention against the existing body of knowledge, known as "prior art." However, the traditional workflow for novelty assessment is heavily dependent on the knowledge and experience of human experts. This model is facing unprecedented pressure. Firstly, subjectivity and inconsistency are inherent flaws. Different examiners may have varying interpretations of "non-obviousness"

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. *GAIB 2025, Hongkong, China*

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1602-7/2025/08

<https://doi.org/10.1145/3766918.3766939>

for the same technical solution, leading to fluctuations in evaluation outcomes. Secondly, the efficiency bottleneck is becoming increasingly severe. With the exponential growth of global research output, the number of patent applications has skyrocketed, forcing examiners to search and compare within a vast ocean of literature and patent databases. This is not only time-consuming and labor-intensive but also prone to errors and omissions.

In recent years, generative AI technologies, led by Large Language Models (LLMs), have achieved breakthrough progress, demonstrating immense potential across various industries. In the scientific domain, LLMs have been successfully applied to assist in generating inventions [2], automatically drafting scientific papers [3], and even participating in the peer-review process. Their powerful natural language understanding, and generation capabilities have opened the door to automating complex text-based tasks. This naturally raises a critical question: can we leverage LLMs to revolutionize the patent novelty assessment process, enhancing its objectivity, consistency, and efficiency?

Despite the promising outlook, directly applying existing LLMs to this task is fraught with difficulties. Mainstream LLMs predominantly rely on an implicit reasoning mode. They operate like a "black box," jumping directly from input to output without revealing the intermediate logical steps [4]. This mode heavily depends on the vast knowledge and statistical patterns learned from training data, rather than on rigorous, step-by-step logical analysis. When applied to patent novelty assessment, which demands precise and structured analysis, the drawbacks of implicit reasoning become glaringly apparent: models are prone to "hallucination," fabricating facts or making exaggerated claims, and their discrimination capability is weak, making it difficult to distinguish between highly similar patents with critical technical differences [5], [6]. These deficiencies lead to unreliable and often incorrect evaluation results, severely undermining the practical utility of LLM-based assessments in serious legal and technical contexts.

Overcoming these challenges necessitates innovative solutions. The academic and industrial communities are actively exploring new approaches. Explicit Chain-of-Thought (CoT) reasoning has been shown to be an effective method for improving the accuracy and interpretability of LLM reasoning [7]. By prompting the model to "think out loud" and explicitly outline its reasoning steps, CoT not only makes the decision-making process transparent but also forces the model to adhere to a more rigorous logical path. Furthermore, multi-agent systems and consensus methods offer a new avenue for enhancing system reliability. By integrating the perspectives of multiple independent agents, these systems can effectively mitigate the biases and errors of any single agent, leading to more robust collective decisions [8].

Building on these insights, we introduce TCL-MFSNE, an advanced multi-agent framework specifically designed for scientific novelty assessment. The contributions of our model are as follows:

- Firstly, we enhance the model's explicit CoT reasoning capabilities through a two-dimensional contrastive learning strategy. Positive-Negative Contrastive Learning: By constructing pairs of authorized (novel) patents and highly similar unauthorized (non-novel) patents, the model learns through

comparison to precisely grasp the "boundary" that distinguishes novel from non-novel. Temporal Contrastive Learning: By situating the patent under evaluation within its historical technological context and comparing it against relevant historical patents, this ensures the temporal accuracy of the novelty judgment, which is critical in patent evaluation.

- Secondly, we incorporate the Byzantine consensus mechanism, originating from the field of distributed computing. This mechanism ensures robustness by requiring a majority agreement among multiple independent agents, effectively tolerating and filtering out "faulty" evaluations from individual agents (e.g., those resulting from hallucination or misjudgment).

In summary, by organically combining explicit CoT reasoning, two-dimensional contrastive learning, and Byzantine consensus, the TCL-MFSNE framework provides a systematic and highly reliable solution for scientific novelty assessment. This paper will elaborate on the design philosophy, technical implementation, experimental validation, and significant advantages of this framework compared to existing methods.

2 Literature Review

Novelty, as an essential criterion in patent evaluations, significantly impacts the determination of patentability, involving assessments against existing knowledge and technological advancements [9]. Historically, novelty evaluations have relied on expert judgments, which often suffer from subjective biases and inconsistencies due to human limitations and interpretation variances [10].

Recent advances in artificial intelligence, particularly LLMs, have demonstrated transformative potential in supporting various scientific creativity tasks, including invention generation, manuscript preparation, and peer reviews [6]. Despite their significant capabilities, LLMs frequently encounter substantial limitations in novelty assessments due to reliance on implicit reasoning processes and susceptibility to hallucinations—generating factually incorrect or exaggerated novelty claims [5], [6].

Implicit reasoning commonly adopted by LLMs is characterized by direct answer generation without intermediate reasoning steps, resulting in evaluations grounded in implicit memory rather than explicit logical deduction [4]. This approach fails in accurately differentiating closely related technological advancements, leading to errors and inaccuracies in novelty evaluations, particularly in contexts requiring careful, explicit differentiation between novel and incremental innovations [3].

Contrastive learning techniques have emerged as promising solutions in NLP for enhancing models' discriminative capabilities. Initially used in computer vision, these methods have effectively addressed discrimination problems in NLP, improving the accuracy of text classification tasks through explicit comparisons between positive and negative examples [11]. Furthermore, incorporating a temporal dimension in contrastive learning, which considers historical context and technological evolution, significantly enhances the model's ability to accurately assess novelty within its historical and technological framework [12].

Multiagent consensus methods further enhance evaluation reliability. Byzantine consensus, originating from distributed computing, provides a robust mechanism to reconcile conflicting agent evaluations by requiring agreement among a majority of agents. This approach effectively mitigates inaccuracies stemming from individual agent errors or inconsistent evaluations, ensuring reliable and stable consensus [13].

Recognizing these advancements and limitations, our research proposes a robust multiagent framework integrating explicit CoT reasoning via positive-negative and temporal contrastive learning combined with Byzantine consensus methods. This comprehensive approach ensures both explicit reasoning accuracy and robust consensus-driven decision-making in scientific novelty evaluations.

3 Methodology

The TCL-MFSNE framework is designed to achieve precise novelty assessment through a structured, multi-stage process. Its overall architecture, depicted in Figure 1, consists of three core, collaborative components: Positive-Negative Contrastive Learning, Temporal Contrastive Learning, and Byzantine Consensus.

3.1 Positive-Negative Contrastive Learning

In this section, we present the details of Positive-Negative Contrastive Learning. The first step in our method is to construct positive-negative pairs that will enable the LLM to learn the distinction between novel and non-novel patents. We begin by selecting a cut-off date, ensuring that the training samples consist only of authorized patents (APs) before this date. This ensures that all the examples used for training are grounded in historically recognized patents.

For each authorized patent (AP_i), we have its application month information T_i . From this, we can retrieve unauthorized patents (UAP) that were filed in the same month, denoted as $UAP^{T_i} = \{UAP_j, UAP_{j+1}, \dots, UAP_{j+N}\}$. These UAPs are the non-novel patents we will use as negative samples for contrastive learning.

To ensure the quality of the training pairs, we use a pretrained language model (PLM) to obtain embeddings for both the authorized patent AP_i and the unauthorized patents UAP_j . We then calculate the cosine similarity between the embeddings of AP_i and all UAP_j in UAP^{T_i} . To improve the training quality, we only retain pairs with a cosine similarity score ≥ 0.8 . This threshold ensures that we are selecting semantically similar patents—those that share relevant content or technological features—while discarding less meaningful pairs. Retaining only highly similar pairs reduces noise in the training process and ensures that the LLM is learning from meaningful contrasts.

Building the Training Pair Database (TDB). Once we have filtered the relevant positive-negative pairs, we store them in the TDB. For each pair, the database contains: Content of both the positive (AP) and negative (UAP) patents; Time information indicating the filing month of the patents; Embedding of the AP, which serves as the representation for the training pair. This database serves as the foundation for positive-negative contrastive learning, where the LLM will learn to distinguish between novel and non-novel patents based on the features embedded in the authorized and unauthorized patent pairs.

For each test patent (tp_k), we follow a similar procedure as for the training data. We first use the same PLM to embed the test patent and then compute the cosine similarity between the test patent’s embedding and all the training pairs in the TDB. By retrieving the most relevant training pairs, the LLM gains insight into how similar or dissimilar the test patent is to the historical positive-negative examples, thereby helping it assess the novelty of the test patent. This step is critical because it allows the LLM to draw parallels between the test patent and previously seen examples. It enhances the model’s ability to reason step-by-step, recognizing patterns of innovation by referring to past examples of novel and non-novel patents.

3.2 Temporal Contrastive Learning

Novelty judgment in patent law has a strict temporal anchor: the “filing date.” A technology’s novelty must and can only be compared against the “prior art” that existed before its filing date. To accurately replicate this legal requirement in our model’s evaluation, we designed the Temporal Contrastive Learning component, centered around the construction of a comprehensive historical patent Vector Database (VDB).

The construction and application of the VDB are as follows:

Historical Data Ingestion: We use all granted patents from our training set (i.e., successful innovations from before the cut-off date) as the knowledge source for “prior art.”

Fine-grained Segmentation and Vectorization: For more precise semantic matching, we do not treat the entire patent as a single unit. Instead, we break it down into smaller paragraphs (e.g., each containing a few sentences or a complete technical point). We then use PLMs to convert these paragraphs into vector embeddings and store them in the VDB. Each vector is tagged with metadata, including its patent’s filing date.

Dynamic Context Retrieval: When a test patent enters the system for evaluation, we first segment and vectorize its content in the same manner. We then use these vectors as queries to perform efficient similarity searches in the VDB (e.g., using libraries like FAISS for approximate nearest neighbor search).

Time-Window Filtering: From the search results, we only keep historical patent paragraphs whose filing date is earlier than the filing date of the patent being evaluated. This step is key to achieving “temporal contrast,” ensuring that all reference information provided to the model strictly adheres to the definition of “prior art.”

Context Injection and Evaluation: The retrieved, time-filtered, relevant technical paragraphs are integrated into the LLM agent’s prompt, serving as the “background knowledge” or “prior art reference” for its novelty judgment. The prompt will guide the model: “Given the patent for evaluation X (filed on date Y), and the following relevant technical descriptions published before date Y, please analyze whether the technical solution of patent X presents a substantive difference and a significant advancement compared to this prior art.”

In this way, Temporal Contrastive Learning ensures that every evaluation is conducted within the correct historical coordinate system, effectively preventing the “anachronistic” error of using future technology to invalidate a past invention.

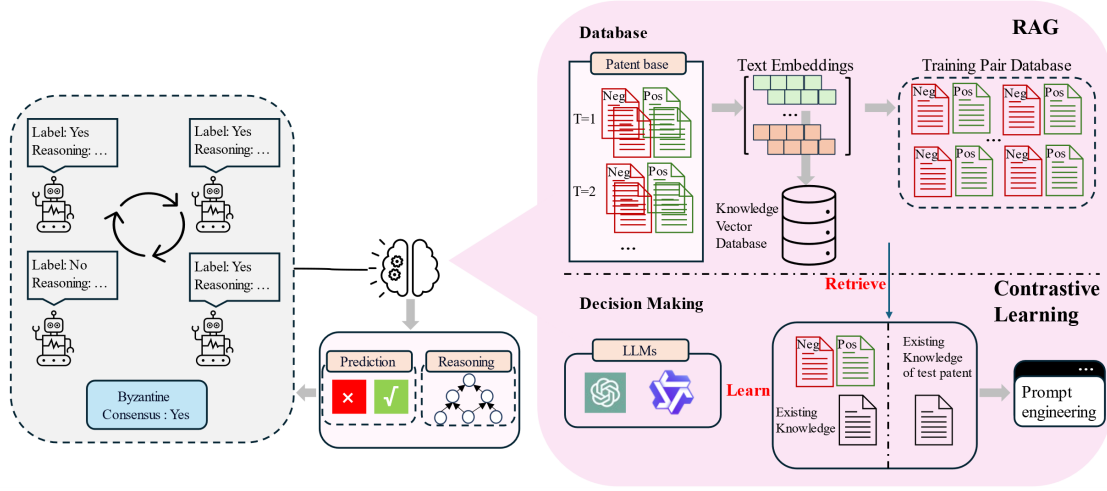


Figure 1: Overview of PatentAgent

3.3 Multi-agent Evaluation with Byzantine Consensus

To overcome the randomness, biases, or model-specific “blind spots” of a single LLM, we introduce a multi-agent collaboration and Byzantine consensus mechanism to pursue robustness and reliability in the evaluation results.

The mechanism operates as follows:

Agent Cluster Configuration: According to Byzantine fault tolerance theory, a system needs at least $3p + 1$ total nodes to tolerate p faulty nodes. In our TCL-MFSNE framework, we configure 4 independent LLM agents ($n=4$), which means the system can tolerate $p=1$ agent failing or providing a completely wrong judgment. To further enhance robustness, we employ a heterogeneous model strategy: two agents use OpenAI’s GPT-4o-mini model, and the other two use Alibaba Cloud’s Qwen-plus model. This diversity in model origins effectively reduces the risk of a collective misjudgment caused by a systemic flaw in a single model family.

Independent Parallel Evaluation: For the same patent under evaluation, all 4 agents receive the identical contextual information generated by the Temporal Contrastive Learning module and are prompted to perform an independent evaluation using the explicit CoT mode trained via Positive-Negative Contrastive Learning. Each agent outputs its judgment (“Novel” or “Not Novel”) along with a detailed reasoning process.

Consensus Decision: After collecting the evaluation results from all agents, the consensus protocol is initiated. We employ a strict majority rule:

If at least 3 agents (i.e., more than two-thirds of the total) reach an agreement (e.g., 3 or 4 judge it as “Novel”), the system outputs that consensus conclusion.

If there is a 2-vs-2 tie (e.g., 2 judge “Novel” and 2 judge “Not Novel”), the system’s output is “Uncertain” or “Requires Human Review.” This indicates that the novelty of the patent is ambiguous, or its technical complexity exceeds the system’s current reliable judgment capability, thus requiring the intervention of a human expert.

Through Byzantine consensus, TCL-MFSNE not only effectively filters out the occasional errors of individual agents but also identifies genuinely contentious cases, thereby maintaining a high level of automation efficiency while providing an interface for collaboration with human experts.

Figure 1 illustrates the TCL-MFSNE architecture, highlighting the systematic interactions between positive-negative contrastive learning, temporal contrastive learning, and Byzantine consensus, which collectively ensure accurate and robust evaluations of patent novelty.

4 Experiments

To validate the effectiveness of the TCL-MFSNE framework, we designed a series of rigorous experiments and compared its performance against several baseline models.

4.1 Experimental Setup

Dataset: In this study, we utilize a dataset sourced from the Patent Review and Adjudication Office (PRAO), which contains both authorized and unauthorized patents from May 2008 to March 2024. The dataset is designed to aid in the evaluation of patent innovation, particularly in distinguishing between patents that are considered novel (authorized) and those that are not (unauthorized). The dataset includes a diverse set of patents across various domains, offering a comprehensive base for training and testing our model.

To build the training and test sets, we use January 1, 2021 as the cut-off date, which separates patents into two categories: training data (patents filed before the cut-off date) and test data (patents filed on or after the cut-off date). This split ensures that the model learns from historical data and is tested on patents that were evaluated after the cut-off date, simulating real-world conditions where new patents are reviewed based on existing knowledge.

The final dataset consists of 8,579 training pairs, which include both authorized (novel) and unauthorized (non-novel) patents; 2,262 positive (authorized) test patents and 1,075 negative (unauthorized)

test patents. The authorized patents are considered as "true innovations," while the unauthorized patents are deemed non-novel for the purpose of this study. These test patents are used to evaluate the model's ability to classify patents correctly as novel or non-novel based on the training it has received.

Experimental Environment: The four LLM agents in the TCL-MFSNE framework were composed of two different advanced models: two ChatGPT-4o-mini (from OpenAI) and two Qwen-plus (from Alibaba Cloud). All models used standard hyperparameter settings, with a temperature of 0.7 to strike a balance between response diversity and logical consistency.

Evaluation Metrics: To evaluate the performance of our framework, we use the following standard metrics:

Accuracy: The proportion of correct predictions (both true positives and true negatives) out of all predictions.

Precision: The proportion of true positives among all instances classified as positive (authorized).

Recall: The proportion of true positives among all actual positives (authorized patents).

F1-score: The harmonic means of precision and recall, providing a balanced measure of the model's performance.

These metrics are calculated based on the confusion matrix, which is derived from the following definitions:

- **True Positives (TP):** Authorized patents correctly classified as authorized.
- **True Negatives (TN):** Unauthorized patents correctly classified as unauthorized.
- **False Positives (FP):** Unauthorized patents incorrectly classified as authorized.
- **False Negatives (FN):** Authorized patents incorrectly classified as unauthorized.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Baseline Models and Ablation Studies

To thoroughly analyze the contribution of each component within the TCL-MFSNE framework, we set up the following comparison models:

Implicit LLM: A baseline model using a single ChatGPT-4o-mini agent with a traditional implicit reasoning prompt (e.g., "Determine if the following patent is novel."), without any contrastive learning or CoT.

Single Agent: A single, fully configured agent from TCL-MFSNE (i.e., trained with two-dimensional contrastive learning), but without the multi-agent consensus. Its individual output is taken as the final result.

Without Positive-Negative CL: An ablation variant of TCL-MFSNE where the positive-negative contrastive learning module is removed. The agents are not trained with the TDB and rely solely on the context provided by temporal contrastive learning.

Without Temporal CL: Another ablation variant of TCL-MFSNE where the temporal contrastive learning module is removed. The agents perform evaluations without any historical prior art context.

4.2 Results

From these results, we can draw several key conclusions:

Superior Performance of TCL-MFSNE: Our TCL-MFSNE framework demonstrates excellent and balanced performance across all four core metrics, achieving an accuracy and F1-score of 85%. This confirms that our proposed integrated design—combining explicit reasoning, two-dimensional contrastive learning, and Byzantine consensus—is highly effective at accurately distinguishing between novel and non-novel patents.

Explicit Reasoning is Far Superior to Implicit Reasoning: Compared to the Implicit LLM, TCL-MFSNE shows a 6-percentage-point improvement in accuracy (from 79% to 85%). Interestingly, the Implicit LLM has very high precision (90%) but low recall (79%). This reveals the behavioral pattern of implicit reasoning models: they are very "conservative," only daring to classify a patent as "novel" when the innovation is extremely obvious, hence the high precision. However, for innovations that are highly similar to prior art and require careful discernment, they tend to classify them as "non-novel," causing many true innovations to be missed (low recall). TCL-MFSNE, through explicit reasoning and contrastive learning, significantly improves recall while maintaining high precision, achieving a much better balance.

Robustness Gains from Byzantine Consensus: The Single Agent model achieved an accuracy of 81%, which is 4 percentage points lower than TCL-MFSNE. This gap clearly quantifies the performance boost provided by the Byzantine consensus mechanism. It shows that through multi-agent voting, the system can effectively correct judgmental biases and random errors of a single agent, making the final result more stable and reliable.

The Critical Role of Positive-Negative Contrastive Learning: After removing positive-negative contrastive learning (Without Positive-Negative CL), the model's accuracy plummeted to 78%, the lowest among all models. Notably, its recall soared to 91%, but its precision was only 80%. This "high recall, low precision" combination indicates that without training on differential pairs, the model becomes "overly optimistic," tending to classify most patents as novel. While it finds almost all truly novel patents (high recall), it also incorrectly labels many non-novel patents as novel (low precision). This proves that positive-negative contrastive learning is crucial for cultivating the model's precise discrimination ability.

The Necessity of Temporal Contrastive Learning: When temporal contrastive learning was removed (Without Temporal CL), the model's accuracy dropped to 81%. Its performance profile, similar to the Implicit LLM, showed "high precision, low recall." This suggests that without historical context, the model cannot accurately understand the boundary of "prior art." When faced with an incremental innovation built upon existing technology, the model might misclassify it as non-novel because it cannot find a sufficiently radical innovation. This highlights that temporal contrastive learning is indispensable for providing the correct evaluation baseline and historical context.

Table 1: Experiment Result

Method	Accuracy	Precision	Recall	F1-Score
patentagent	0.85	0.86	0.85	0.85
Implicit LLM	0.79	0.9	0.79	0.84
Single Agent	0.81	0.83	0.81	0.82
Without Positive-Negative Contrastive Learning	0.78	0.8	0.91	0.85
Without Temporal Contrastive Learning	0.81	0.9	0.81	0.85

In summary, the experimental results robustly demonstrate the integrity and synergistic effect of the TCL-MFSNE framework’s design. Each component targets a specific weakness of LLMs in patent assessment, and their organic combination ultimately builds an advanced evaluation system that surpasses existing methods in accuracy, robustness, and overall performance.

5 Conclusion

The assessment of novelty is a fundamental aspect of various fields, particularly in patent law, where it plays a pivotal role in determining whether an invention qualifies for intellectual property protection. This process is not only crucial for securing patent rights but also serves as a public record of technological advancements, contributing to the ongoing development of industries and research fields. For patent offices, novelty assessment is a key decision-making criterion that influences whether a patent application will be granted. Given the increasing volume of patent filings and the complexity of technological innovations, ensuring accurate and reliable evaluations of novelty is essential. Traditional patent evaluation methods often rely on expert judgment, which can be time-consuming and prone to subjectivity. With the rise of LLMs, there is an opportunity to automate and enhance this process. However, existing LLM-based approaches for novelty assessment often rely on implicit reasoning abilities and lack valid evaluation frameworks to support their reliability and accuracy.

To address these challenges, we developed the PatentAgent framework, a novel approach that integrates explicit contrastive learning techniques and Byzantine consensus to improve the novelty assessment of patents. Our framework incorporates two key components: positive-negative contrastive learning, which trains the LLM to differentiate between novel and non-novel patents, and temporal contrastive learning, which situates patent evaluations within the broader context of historical technological advancements. These techniques guide the LLM through a more structured reasoning process, enabling it to assess the novelty of a patent relative to

existing knowledge. Furthermore, we integrate a Byzantine consensus mechanism to ensure the stability and consistency of the final evaluation, enhancing the reliability of the model’s decision-making process. We tested our framework using a comprehensive dataset of authorized and unauthorized patents, with results demonstrating that PatentAgent outperforms baseline models in key performance metrics, such as accuracy and F1 score. Specifically, PatentAgent achieved an accuracy of 85%, surpassing the performance of other approaches, including the Implicit LLM and single-agent models.

References

- [1] J. H. Ashtor, “Modeling patent clarity,” *Research Policy*, vol. 51, no. 2, p. 104415, Mar. 2022, doi: 10.1016/j.respol.2021.104415.
- [2] R. Abbott, “Allow patents on AI-generated inventions — for the good of science,” *Nature*, vol. 620, no. 7975, pp. 699–699, Aug. 2023, doi: 10.1038/d41586-023-02598-2.
- [3] T. Ifargan, L. Hafner, M. Kern, O. Alcalay, and R. Kishony, “Autonomous LLM-driven research from data to human-verifiable research papers,” Apr. 2024. Accessed: May 21, 2024. [Online]. Available: <https://www.semanticscholar.org/paper/Autonomous-LLM-driven-research-from-data-to-papers-Ifargan-Hafner/e4eb81ad222ba047770d5a90bdd7406c138c6126>
- [4] Y. Yu, “LLMs Do Not Think Step-by-step In Implicit Reasoning,” Nov. 24, 2024, *arXiv*: arXiv:2411.15862, doi: 10.48550/arXiv.2411.15862.
- [5] Z. Ji et al., “Survey of Hallucination in Natural Language Generation,” *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, Dec. 2023, doi: 10.1145/3571730.
- [6] A. Kankanalli, “Peer Review in the Age of Generative AI,” *JAIIS*, vol. 25, no. 1, pp. 76–84, 2024, doi: 10.17705/1jais.00865.
- [7] J. Wei et al., “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” Jan. 10, 2023, *arXiv*: arXiv:2201.11903, doi: 10.48550/arXiv.2201.11903.
- [8] H. Chen, W. Ji, L. Xu, and S. Zhao, “Multi-Agent Consensus Seeking via Large Language Models,” Oct. 31, 2023, *arXiv*: arXiv:2310.20151, doi: 10.48550/arXiv.2310.20151.
- [9] Z. Iqbal and S. Sadaf, “Artificial Intelligence / Machine Learning-based Innovations – A Review of Patent Eligibility Standards, Policies, Open Issues and Guiding Framework,” *Expert Systems with Applications*, vol. 239, p. 121819, Apr. 2024, doi: 10.1016/j.eswa.2023.121819.
- [10] H. B. De Vries and T. I. Lubart, “Scientific Creativity: Divergent and Convergent Thinking and the Impact of Culture,” *Journal of Creative Behavior*, vol. 53, no. 2, pp. 145–155, Jun. 2019, doi: 10.1002/jocb.184.
- [11] J. Chen, R. Zhang, Y. Mao, and J. Xu, “ContrastNet: A Contrastive Learning Framework for Few-Shot Text Classification,” May 16, 2023, *arXiv*: arXiv:2305.09269, doi: 10.48550/arXiv.2305.09269.
- [12] D. Encaoua, D. Guellec, and C. Martínez, “Patent systems for encouraging innovation: Lessons from economic analysis,” *Research Policy*, vol. 35, no. 9, pp. 1423–1440, Nov. 2006, doi: 10.1016/j.respol.2006.07.004.
- [13] M. Castro and B. Liskov, “Practical Byzantine Fault Tolerance,” 1999.